TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Smart 3D GelMA-Based Functional Biomaterials with Broad Applicability and Industrial Scalability			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio · Medical		Bio · Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Bio - Biometerials	Industrial Biotechnology	Functional Biomaterials
Concept of Proposed Technology	The proposed technology focuses on the development and scalable production of smart hydrogel biomaterials based on gelatin methacryloyl (GelMA), a photocrosslinkable derivative of naturally derived gelatin [1]. The technology addresses current limitations of conventional biomaterials by enabling precise control of sol–gel transition mechanisms, mechanical properties, and biological functionality, while remaining cost-effective and suitable for mass production [2]. GelMA-based hydrogels retain the intrinsic advantages of gelatin, including excellent biocompatibility, cell-adhesive RGD sequences, and enzymatically controllable biodegradation, while overcoming limitations such as low mechanical strength and poor structural stability through photocrosslinking chemistry [3]. The proposed technology enables the production of highly tunable 3D hydrogel systems suitable for applications in tissue engineering, regenerative medicine, 3D bioprinting, injectable medical products, and in vitro disease models. In addition, the technology incorporates a pH automatic adjustment system for synthesis optimization, enabling			

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	environmentally friendly, reproducible, and versatile mass production of GelMA and related methacrylated biomaterials.
Rationale for Joint R&D Project	The demand for medical biomaterials is rapidly increasing due to population aging, advances in regenerative medicine, and growing interest in precision healthcare [4]. However, despite extensive research activity and strong academic interest as well as growing industrial demand, many high-performance biomaterials—particularly GelMA-based bioinks and hydrogels—remain heavily dependent on imports, limiting domestic supply security and increasing costs [5]. Small and medium-sized enterprises face substantial challenges in entering the biomaterials market due to high production costs, limited scalability, and barriers to commercialization. While the need for functional, biocompatible, and affordable biomaterials is widely recognized, practical implementation and industrialization remain insufficient. This joint R&D project is required to: • Establish scalable and standardized GelMA production technologies • Reduce production costs while maintaining high biological and mechanical performance • Enable industrial adoption and localization of key biomaterials • Accelerate commercialization through international collaboration This joint R&D project is particularly relevant for Korean SMEs developing bioinks and hydrogel-based biomaterials for 3D bioprinting, regenerative medicine, and cell culture applications. Such SMEs face challenges related to the stable supply, reproducibility, and cost of GelMA, which is a core component of many bioink formulations. Through international collaboration, this project aims to establish an industrially scalable GelMA production platform that directly addresses SME needs in bioink manufacturing and enables competitive product development for domestic and global markets (**Figure 1**).
Key Words	1. Gelatin Methacryloyl (GelMA) 2. Functional Biomaterials 3. Smart Hydrogels 4. Scalable Production 5. 3D Bioprinting

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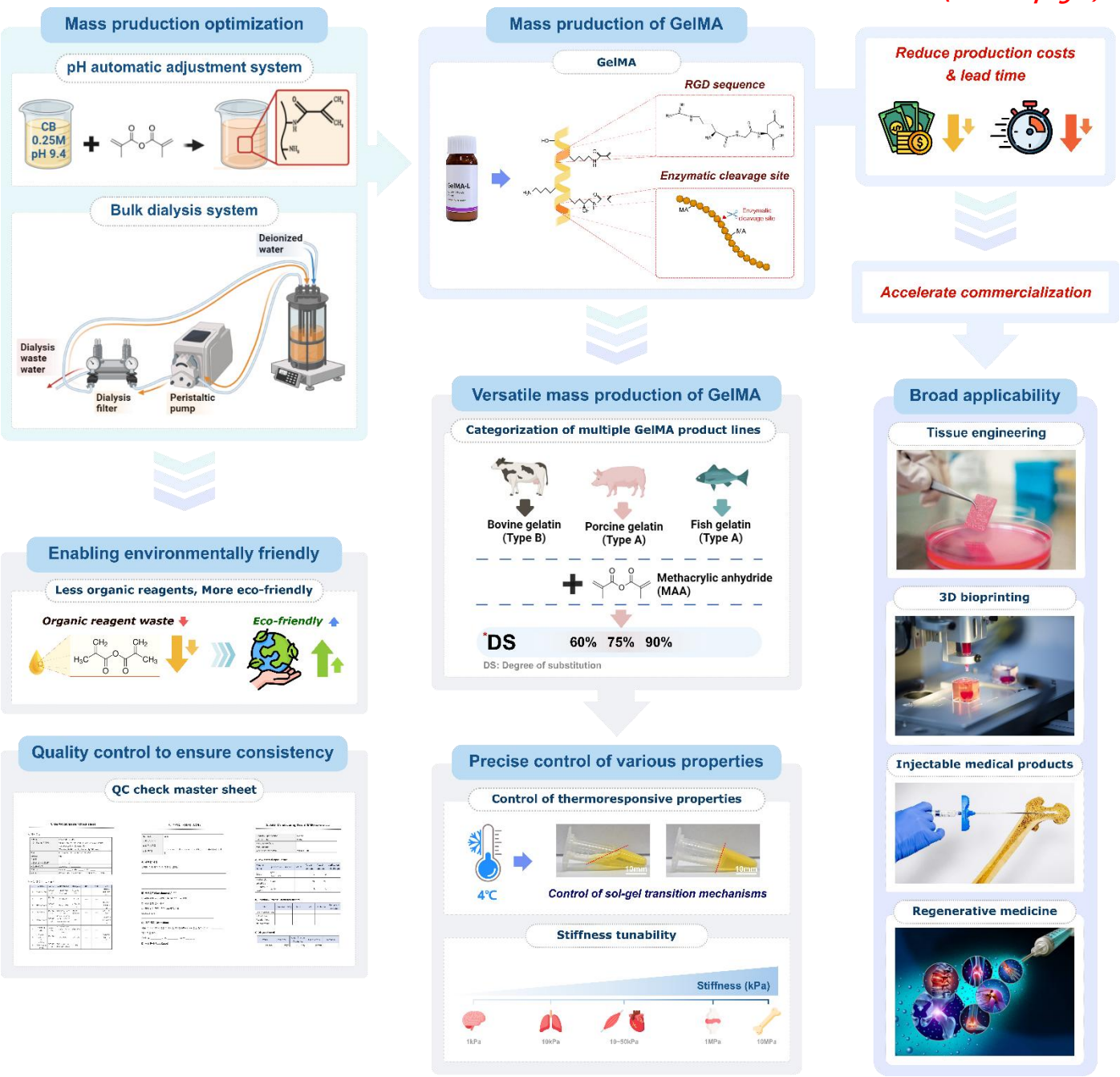

Figure 1. Technology overview. Mass production, optimization, and application of smart GelMA biomaterials.

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (TRL 3–4)	End of the Project	Level (TRL6)
R&D Objectives	The objectives of this joint R&D project are to: • Develop a robust and scalable GelMA synthesis process suitable for industrial production • Establish quantitative control of mechanical, chemical, and biological properties • Optimize environmentally friendly production routes using minimal organic solvents • Validate pilot-scale GelMA production for biomedical applications • Prepare the technology for commercialization and downstream product development, including application-specific GelMA formulations suitable for bioink production and cell-compatible hydrogel systems.			
R&D Target Specifications	• Controlled degree of methacryloyl substitution with defined tolerance ranges • Tunable mechanical properties (e.g., stiffness and viscoelasticity) based on application needs • High batch-to-batch reproducibility in physicochemical properties • Low residual impurities and reduced hazardous waste generation • Pilot-scale production feasibility with cost reduction compared to imported materials			
R&D Plan	**Phase 1: Material and Process Optimization (Months 1–6)** • Optimization of GelMA synthesis parameters • Implementation of pH automatic adjustment system • Laboratory-scale reproducibility and characterization **Phase 2: Scale-Up and Pilot Production (Months 7–18)** • Scale-up of synthesis and purification processes • Pilot-scale GelMA production • Mechanical, chemical, and biological performance evaluation **Phase 3: Application Validation and Commercial Readiness (Months 19–24)** • Application-oriented testing with a focus on bioink formulation, printability, and hydrogel performance under SME-relevant manufacturing conditions • Manufacturing feasibility assessment with SME partner • Preparation for technology transfer and commercialization **Figure 2**			

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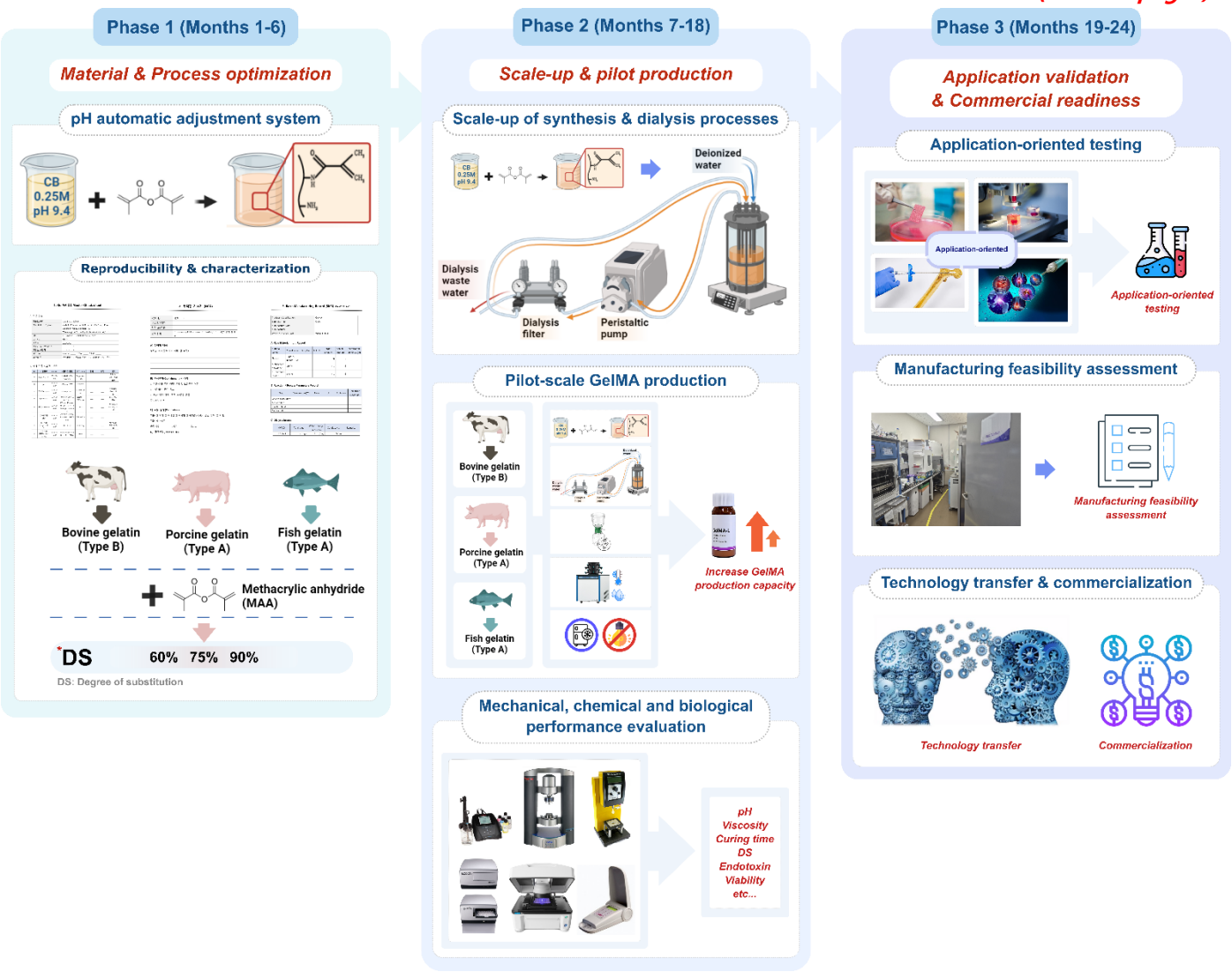

Figure 2. Overall R&D plan. Phase 1 will encompass material and process optimization. In Phase 2, smart hydrogel production scale will be upgraded according to production protocol. In Phase 3, Validation will be performed for the commercial applications.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead GelMA material design, synthesis optimization, and characterization • Develop and validate pH-controlled and environmentally friendly production processes • Define key quality attributes and performance metrics • Provide technical expertise, research infrastructure, and joint IP contributions This proposed technology directly aligns with the mission and expertise of the Fraunhofer Institute for Manufacturing Engineering and Automation IPA, particularly within the Pharma and Medical Industry department and the Intelligent Systems and Production group. The project focuses on translating advanced biomaterials from laboratory-scale synthesis to scalable, automated, and quality-controlled manufacturing processes. Fraunhofer IPA contributes strong competencies in process engineering, intelligent production systems, pilot-scale manufacturing, and technology transfer for regulated medical and pharmaceutical applications. The development of standardized and reproducible GelMA production platforms, including automated pH-controlled synthesis and environmentally friendly process design, represents a core application area of Fraunhofer IPA's production-oriented R&D approach, enabling efficient industrial adoption by SMEs.
Expected R&R of Korean SME Partner	• Lead scale-up, pilot manufacturing, and process engineering • Provide facilities for pilot-scale production and validation • Conduct application-specific testing and market-oriented evaluation • Support commercialization strategy and regulatory alignment The Korean SME partner is expected to be a company specializing in the development and manufacturing of bioinks or hydrogel-based biomaterials for 3D bioprinting, regenerative medicine, or advanced cell culture applications. The SME will contribute expertise in pilot-scale production, formulation development, and application-oriented performance evaluation. The SME will adapt the scalable GelMA production process to its specific bioink requirements, conduct validation under practical manufacturing conditions, and support downstream commercialization, including market deployment and customer engagement.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver: • A cost-effective, scalable GelMA production platform suitable for industrial use • Reduced dependence on imported biomaterials and enhanced domestic supply capability • New commercialization opportunities for Korean SMEs in regenerative medicine, bioprinting, and injectable biomaterials • Long-term economic and industrial impact through technology localization, joint IP creation, and international collaboration The project will contribute directly to Korea's Advanced Bio strategic technology field and support the growth of high value-added biomaterials industries. The project will lower entry barriers for Korean SMEs by reducing dependence on imported biomaterials and providing access to standardized, cost-effective GelMA production technologies, thereby strengthening domestic supply chains and enhancing global competitiveness. The project will support Korean bioink and biomaterials SMEs by providing access to a standardized and cost-effective GelMA supply, reducing dependence on imported raw materials, and enabling consistent product quality. This will accelerate commercialization of bioinks and hydrogel products for research, industrial, and future clinical applications.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization